MILITARY SPECIFICATION
CUPOLA, FLAME GUN, M8

1. SCOPE

1.1 This specification covers a cupola for use in the M10-8 flame thrower.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat.
SS-G-659	-	Graphite; Lubricating, Flake.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape; Pressure-Sensitive Adhesive, Waterproof - For Packaging and Sealing.

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed, and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	-	Treatment and Painting of Material.
MIL-P-3542	-	Primer, Pressure-Sensitive Tape.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-C-7769	-	Cushioning Material, Bound Fiber.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-V-13811	-	Varnish, Waterproofing, Electrical, Ignition.
MIL-C-17435	-	Cushioning Material, Fibrous Glass.
MIL-C-40010	-	Cushioning Material, Packaging, Polyvinyl Chloride, Plasticized, Cellular.
MIL-F-51156	-	Flame Thrower, Mechanized, Main Armament, APC-Mounted, M10-8.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

DL 81-1-9100 - Cupola, Flame Gun, M8.
(All drawings referenced in this specification are listed on Dwg DL 81-1-9100.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL 81-1-9100 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the cupola shall conform to the specifications and drawings listed on Drawing DL 81-1-9100 and subsidiary drawings.

3.2 Assembly. The cupola shall be assembled as shown on Drawing E81-1-9100.

3.3 Electrical circuits.

3.3.1 Insulation resistance. The wire shall show an open circuit when 500 volts of direct current is applied and tested as specified in 4.6.1.

3.3.2 Continuity. The wire circuit shall be continuous and register a maximum of 1 ohm when tested as specified in 4.6.2.

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3.3.3 Components. The following components shall meet the requirements of 3.3.1 and 3.3.2.

(a)	Cable, special purpose	(Dwg B81-1-9058).
(b)	Electrical brush	(Dwg C81-1-9083).
(c)	Cable, electrical	(Dwg C81-1-9297).
(d)	Cable, electrical	(Dwg C81-1-9298).
(e)	Cable, electrical	(Dwg C81-1-9349).
(f)	Slip ring, electrical	(Dwg D81-1-9079).
(g)	Cable, electrical	(Dwg D81-1-9314).

Continuity of all circuits shall be indicated at all times through 360 degrees rotation.

3.4. Leakage. The following components shall not leak when subjected to a hydrostatic pressure of 550 ± 10 pounds per square inch gage (psig) for a minimum period of 1 minute when tested as specified in 4.6.3.

(a)	Cap, pressure	(Dwg B81-1-9114).
(b)	Cap, filler	(Dwg C81-1-9115).
(c)	Hose, medium pressure	(Dwg B81-1-9330).
(d)	Fitting, adapter	(Dwg B81-1-9335).
(e)	Nozzle, flame gun	(Dwg E81-1-7476).
(f)	Body, flame gun	(Dwg E81-1-7481).
(g)	Cylinder	(Dwg D81-1-7490).
(h)	Inlet piping	(Dwg E81-1-9210).
(i)	Elbow, trunnion	(Dwg E81-1-9218).

3.5. Plastic handle. The handle connected to a 24 volt direct current power source shall function properly when tested as specified in 4.6.4.

3.6 Accumulator (Dwg C81-1-9265).

3.6.1 Body. The accumulator body shall not leak when subjected to a hydrostatic pressure of 1000 ± 20 psig for a minimum period of 1 minute when tested as specified in 4.6.5.1.

3.6.2 Bag. The bag shall not leak more than 5 pounds in 5 hours, subjected to a pressure of 160 ± 5 psig when tested as specified in 4.6.5.2.

3.6.3 Endurance. The accumulator shall function at a temperature of -40° and 125°F. Pressurize the bag to 160 ± 5 psig and the body to 325 ± 5 psig. Increase the pressure in the body to 950 ± 10 psig and then reduce it to 325 ± 5 psig, this constitutes 1 cycle when tested as specified in 4.6.5.3.

3.7 Valves (Dwg C81-1-9053 and C81-1-9055).

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3.7.1 Leakage. The valve shall not leak when subjected to a hydrostatic pressure of 550 ± 10 psig for a minimum period of 1 minute. The valve shall then be pressurized with air at 325 ± 5 psig against the closed port and shall not leak more than 15 psig in a minimum period of 5 hours when tested as specified in 4.6.6.1.

3.7.2 Endurance. The valve shall function at a temperature range of -40° to 125°F. The closed port shall be pressurized with 325 ± 5 psig of air, and the solenoid shall be energized for 5 ± 1 seconds and shall then be released. This consitiutes 1 cycle when tested as specified in 4.6.6.2.

3.8 Valve, pneumatic (Dwg C81-1-9156).

3.8.1 Leakage. The valve shall not leak when subjected to 550 ± 10 psig air pressure for a minimum period of 1 minute. The valve shall then be pressurized to 325 ± 5 psig and shall not leak more than 15 psig when tested as specified in 4.6.7.1.

3.8.2 Endurance. The valve shall function at a temperature range of -40° to 125°F in accordance with table I. With valve closed, the solenoid shall be energized to open valve. It shall be kept opened for 1 second and then the solenoid shall be de-energized to close valve. This constitutes 1 cycle when tested as specified in 4.6.7.2.

Table I. Operating conditions

Voltage direct current	Pressure	Opening time (full open) milliseconds (max)	Change time to fully close milliseconds (max)
18 to 30	325 - 5 psig	300	400

3.9 Fire control box (Dwg D81-1-9251). All switches shall function using 24 volt direct current. The elapsed time indicator switch when engaged and released shall indicate the following time intervals in seconds in the following order: 3 - 6 - 7 - 5 - 6 - 12 - 3 - 7 - 6 - 5. The total indicated time shall not differ from that on the calibrated timer by more than 1 second; the fuel warning light shall light at the 10 second mark and remain lit until the pointer reaches the zero position when tested as specified in 4.6.8. The control box shall be tested at -25° ± 5°F, 70° ± 5°F and 125° ± 5°F.

3.9.1 Insulation resistance. The wire in the fire control box shall conform to 3.3.1 when tested as specified in 4.6.1.

3.9.2 Continuity. The wire circuit shall conform to 3.3.2 when tested as specified in 4.6.2.

3.10 Rotary joint (Dwg E81-1-9076).

3.10.1 Elbow leakage. The elbow (Dwg E81-1-7462) shall not leak when subjected to a hydrostatic pressure of 550 ± 10 psig for 1 minute when tested as specified in 4.6.9.1.

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3.10.2 Functioning. The assembly shall be capable of manually rotating 360 degrees without the aid of any mechanical devices when tested as specified in 4.6.9.2.

3.10.3 Insulation resistance. The wire in the joint shall conform to 3.3.1 when tested as specified in 4.6.1.

3.10.4 Continuity. The wire circuit shall conform to 3.3.2 when tested as specified in 4.6.2.

3.11 Gear drive, elevation (Dwg D81-1-9280). The average force applied tangentially to the hand crank of elevation gearbox at a uniform rate to maintain gun movement shall not exceed 10 pounds. No individual reading shall exceed 12 pounds, when tested as specified in 4.6.10.

3.12 Box, gear, azimuth (Dwg D81-1-7459). The average force applied tangentially to the hand crank of azimuth gear box at a uniform rate to maintain cupola movement shall not exceed 12 pounds. At uniform hand crank speeds below 30 revolutions per minute chatter-free traverse shall be obtained when tested as specified in 4.6.11.

3.13 Gasoline tank (Dwg E81-1-9123). The permanent volumetric expansion of the gasoline tank shall not exceed 12 percent of the total volumetric expansion when subjected to an internal water pressure of 550 ± 10 psig for a minimum period of 1 minute after the pressure is stabilized. The tank shall be tested as specified in 4.6.12. Prior to this test the tank shall not be subjected to any pressure greater than 495 psig.

3.14 Ignition exciter (Dwg D81-1-9056). The exciter shall perform as specified in the operating characteristic table on Drawing D81-1-9056, when connected to a source of variable voltage and the specified amperage is applied and tested as specified in 4.6.13.

3.15 Cupola (Dwg E81-1-9100).

3.15.1 Hatch cover leakage. The hatch cover shall not leak when subjected to a spray of water 5 gallons per minute (garden hose) for a minimum period of 1 minute when tested as specified in 4.6.14.1.

3.15.2 Cupola leakage. The cupola assembly shall not leak when the accumulator bag is subjected to an air pressure of 160 ± 5 psig and the fuel system to an air pressure of 550 ± 10 psig for a minimum period of 1 minute, then it shall be reduced to 325 ± 5 psig when tested as specified in 4.6.14.2.

3.15.3 Functioning. The cupola assembly shall be functioned and tested in accordance with Specification MIL-F-51156.

3.16 Workmanship. The cupola assembly shall be free from foreign matter (oil, grease, or dirt) and damage (cracks, dents, or deformation) which could affect functioning.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract of purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratories acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government's responsibility. The Government will be responsible for witnessing the tests in 4.6.5.3, 4.6.6.2 and 4.6.7.2.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will insure equal or better quality assurance by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the components produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4 Sampling for nondestructive tests, (endurance). Sampling shall be conducted in accordance with Standard MIL-STD-105, level S-2.

4.5 Inspection provisions.

4.5.1 Examination. Each item shall be examined in accordance with the list of defects. If any item contains any defect listed in 4.5.3, that item shall be rejected.

4.5.2 Nondestructive tests.

4.5.2.1 Endurance. Sample components shall be tested in accordance with 4.6 as applicable and Standard MIL-STD-105 using an AQL of 2.5 percent defective.

4.5.2.2 Components and cupola assembly. Except for endurance, each component and cupola assembly shall be tested in accordance with 4.6 as applicable. If any item fails any test, that item shall be rejected.

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4.5.3 List of defects.

4.5.3.1 Cupola, flame gun, M8 (Dwg E81-1-9100).

1. Component missing or incorrectly assembled
*2. Spacer mounting hole, diameter incorrect
*3. Spacer mounting hole, location incorrect
4. Traverse lock pin nonfunctioning
*5. Nut sleeve coupling, threads incorrect
6. Hatch cover lock nonfunctioning
7. Hinge lock handle nonfunctioning
8. Damage (kinked line or hose deformation 3.16)
9. Foreign matter (3.16)
10. Protective finish missing or incorrect
11. Tray, ammo not as specified on Drawing D81-1-9181

4.5.3.2 Flame gun (Dwg E81-1-7506).

1. Spring, load height (after seasoning) incorrect (Dwg B81-1-7522)
2. Cotter pin missing (pintle rod and piston assembly)
3. Silver braze or pin missing (Dwg C81-1-7603)
4. "O"-ring lubricant missing or incorrect
5. Protective finish missing or incorrect
6. Insert nozzle, machine finish incorrect (Dwg B81-1-7477)
7. Insert nozzle (1.160) diameter incorrect (Dwg B81-1-7477)
8. Insert nozzle (3.000) blend radius incorrect (Dwg B81-1-7477)
9. Pintle rod, head (1.12) diameter incorrect (Dwg C81-1-7603)
10. Pintle rod, head (32) machine finish incorrect (Dwg C81-1-7603)
11. Straightener flow, taper 70 degrees and 7 minutes incorrect (Dwg C81-1-7486)
12. Straightener flow, (1.38) diameter incorrect (C81-1-7486)
13. Nozzle, flame gun (3.750) blend radius incorrect (Dwg E81-1-7476)
14. Nozzle, flame gun (3.750) blend radius machine finish incorrect (Dwg E81-1-7476)

4.5.3.3 Inlet piping (Dwg E81-1-9210).

*1. Collar, eccentricity excessive
*2. Collar, diameter (4.494) incorrect
*3. Collar, diameter (5.032) incorrect
*4. Collar, diameter (5.3742) incorrect
5. Machine finish incorrect

*For repair parts only.

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4.5.3.4 Housing elbow (Dwg E81-1-9077).

*1. Diameter (4.869) incorrect
*2. Diameter (5.3758) incorrect
3. Machine finish (32) or (63) incorrect

4.5.3.5 Box, gear, azimuth (Dwg D81-1-7459).

1. Gear, hardness incorrect (Dwg B81-1-6293)
2. Gear, hardness incorrect (Dwg B81-1-6294)
3. Gear, harndess incorrect (Dwg C81-1-7455)
4. Pin (MS 171594) loose

4.5.3.6 Spring (Dwg C81-1-9059).

1. Hardness incorrect

4.5.3.7 Preparation for delivery (section 5).

1. Opening not processed as specified
2. Anchoring, bracing, or blocking missing or insufficient to prevent movement of the unit
3. Container closure incorrect
4. Marking incorrect, missing, or illegible

4.6 Tests. Unless otherwise specified components requiring endurance test shall be cycled 400 times each at temperature of 70° ± 5°F, 125° ± 5°F, and -45° ± 5°F. All other testing shall be conducted at ambient temperatures. Unless otherwise specified, hydrostatic testing shall use hydraulic oil conforming to Specification MIL-H-5606.

4.6.1 Insulation resistance. The wire shall be tested using a "Megger Insulation Tester" or similar type tester for conformance to 3.3.1.

4.6.2 Continuity. The wire circuit shall be checked with an ohmmeter for conformance to 3.3.2.

4.6.3 Leakage. The components subjected to the specified pressure and time shall comply with the requirements of 3.4. The caps shall be connected to a quick-disconnet fitting.

4.6.4 Plastic handle. The handle connected to the specified power source and an indicator connected to each of the two circuits shall conform to 3.5 when the trigger in the circuit is depressed.

4.6.5 Accumulator.

*For repair parts only.

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4.6.5.1 Body. The body when subjected to the specified pressure and time shall conform to 3.6.1.

4.6.5.2 Bag. The bag when subjected to the specified pressure and time shall conform to 3.6.2.

4.6.5.3 Endurance. The accumulator when subjected to the specified pressure, temperature, and time (see 4.6) shall conform to 3.6.3. The accumulator shall be cycled in the vertical position at both temperatures.

4.6.6 Valves.

4.6.6.1 Leakage. The valve when subjected to the specified pressure and time shall conform to 3.7.1.

4.6.6.2 Endurance. The valve when subjected to the specified temperatures, pressure, and time shall conform to 3.7.2. The valve shall be cycled in the horizontal position at both temperatures.

4.6.7 Valve.

4.6.7.1 Leakage. The valve when subjected to the specified pressures and times shall conform to 3.8.1.

4.6.7.2 Endurance. The valve when subjected to the pressure and temperatures specified shall conform to 3.8.2. The valve shall be cycled in the horizontal position at a 60 degree elevation with the solenoid forward.

4.6.8 Fire control box. The fire control box when subjected to the specified current shall conform to 3.9. The elapsed time indicator and the calibrated electric timer shall be connected so that they will operate and stop simultaneously.

4.6.9. Rotary joint.

4.6.9.1 Elbow leakage. The elbow when subjected to the specified pressure and time shall conform to 3.10.1.

4.6.9.2 Functioning. The rotary joint shall rotate as specified, easily and smoothly, and conform to 3.10.2.

4.6.10 Gear drive, elevation. The drive shall function and conform to 3.11.

4.6.11 Box, gear, azimuth. The gear box shall function and conform to 3.12.

4.6.12 Gasoline tank. The tank when subjected to the specified pressure and time shall conform to 3.13. The pressure gage shall permit reading to an accuracy of 1 percent, and the expansion gage to an accuracy of either 1 percent or 0.1 cubic centimeter for test.

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4.6.13 Ignition exciter. The exciter when subjected to the specified voltage and amperage shall conform to 3.14.

4.6.14. Cupola.

4.6.14.1 Hatch cover leakage. The hatch cover when subjected to the specified spray and time shall conform to 3.15.1.

4.6.14.2 Cupola leakage. The cupola when subjected to the specified pressure and time shall conform to 3.15.2.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.4)

5.1 Cleaning and drying. The upper and lower cupola assemblies, including component parts thereto, shall be cleaned and dried in conformance with Specification MIL-P-116, using cleaning method C-1 and the most applicable of drying methods listed therein.

5.2 Preservation and packaging. Levels shall be as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Unit preservation and packaging.

5.2.1.1.1 Preservatives and application. Preservatives specified herein, including the application thereof, shall conform to the applicable requirements of Specification MIL-P-116. Preservatives shall not be applied to painted surfaces.

5.2.1.1.2 Tape and tape application. Unless otherwise specified, pressure-sensitive tape shall conform to type III, class 1 of Specification PPP-T-60. Prior to application of tape to painted surfaces, such surfaces shall be coated with pressure-sensitive tape primer, conforming to Specification MIL-P-3542, to increase adhesiveness of the tape and durability of seals.

5.2.1.1.3 Electrical ignition waterproofing varnish. This varnish specified herein shall conform to Specification MIL-V-13811.

5.2.1.1.4 Repainting. Damaged portions of painted areas shall be treated and repainted in accordance with Specification MIL-T-704, type A, except that the color of the finish coat shall conform to the requirements as shown on applicable drawings.

5.2.1.1.5 Upper cupola assembly. Unless otherwise specified, exposed unpainted nonoperating surfaces, including recessed areas shall be coated with type P-1 preservative compounds.

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5.2.1.1.5.1 Cupola. The hatch hinges and hatch hinge locking device shall be lubricated with automotive and artillery grease conforming to Specification MIL-G-10924. After lubrication, wipe off excess and cover grease fittings with snug fitting plastic grease cap covers. The two hatch locks on underside of hatch shall be oiled with type P-10, grade 1 or 2 preservative oil. The interrupter ring shall remain in the standard mounted position on the cupola.

5.2.1.1.5.2 Fuel filling inlet cover. The cover latch and hinge shall be oiled with type P-10, grade 1 or 2 preservative oil.

5.2.1.1.5.3 Fuel filling inlet. The cap shall be removed and the unpainted surfaces of the filler inlet and cap shall be coated with type P-10, grade 2 or 3 preservative. The cap shall be replaced and sealed in place with pressure-sensitive tape, overcoating the tape with electrical ignition water-proofing varnish.

5.2.1.1.5.4 Flame gun. The internal surfaces of the trunnion inlet and flame gun shall be coated by flushing with type P-10, grade 2 or 3 preservative oil.

5.2.1.1.5.5 Elevation drive assembly and elevation handle. The elevation bar shall be retracted into a closed position and locked. The unpainted exposed portions of the elevation bar shall be coated with automotive and artillery grease conforming to Specification MIL-G-10924. The preserved portions shall then be wrapped with grade C grease-proof barrier-material, conforming to Specification MIL-B-121, extending the wrap approximately 2 inches beyond the preserved area. The wrap shall be secured in place with pressure-sensitive tape, over-coating the wrap and tape with electrical ignition waterproofing varnish. The elevation handle shall be secure to the elevation drive assembly with pressure-sensitive tape to prevent rotation.

5.2.1.1.5.6 Traverse drive assembly. Flaked graphite conforming to Specification SS-G-659 shall be applied to the locking device of the gear box nandle, actuating lock during application to insure penetration of the graphite.

5.2.1.1.5.7 Ring gear. The ring-gear (Ord Dwg F8683038) shall be coated with type P-10 grade 2 of 3 preservative oil.

5.2.1.1.6 Lower cupola assembly. Unless otherwise specified, all unpainted exposed areas except for operating mechanisms shall be coated with type P-1 preservative compound.

5.2.1.1.6.1 Inlet piping assembly. The internal surfaces of the inlet piping assembly shall be coated by flushing with type P-10, grade 2 or 3 preservative oil. The opening at the lower end of the inlet piping, next to the rotary joint shall be closed with a metal plug and pipe clamp coupling. The bolts securing the pipe clamp coupling and the exposed surface of the metal plug shall be coated with type P-1 preservative compound.

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5.2.1.1.6.2 Gasoline tank. The gasoline tank shall be completely filled with type P-10, grade 2 or 3 preservative oil and then drained. Unpainted surfaces of the filler inlet, filler neck and cap shall be coated with the same preservative oil, replacing cap and sealing in place with pressure-sensitive tape, overcoating the tape with electrical ignition waterproofing varnish.

5.2.1.1.6.3 Hydraulic accumulator. The accumulator shall be processed in accordance with the manufacturer's instructions, using only the manufacturer's recommended lubricants.

5.2.1.1.6.4 Operator's seat. The threaded spindle on the underside of the seat shall be coated with automotive and artillery grease conforming to Specification MIL-G-10924, after which the excess grease shall be removed.

5.2.1.1.6.5 Rotary joint. The fittings on or around the rotary joint housing shall be covered with tape, overcoating the tape with the electrical ignition waterproofing varnish. Lubrication of the joint shall be in accordance with the manufacturer's instructions, using only the manufacturer's recommended lubricants.

5.2.1.1.7 Electrical components. No petroleum type preservatives shall be applied to the electrical components.

5.2.1.1.7.1 Fire control and ignition exciter. The fire control box and ignition exciter shall be wrapped in place with layers of neutral nonhygroscopic cushioning material, conforming to Specification PPP-C-843, MIL-C-7769, MIL-C-17435, or MIL-C-40010 of sufficient quantity to protect projecting switches, lights, and dials. The cushioning shall be secured with pressure-sensitive tape. This wrap shall then be overwrapped with grade C, greaseproof barrier material conforming to Specification MIL-B-121, securing the overwrap with pressure-sensitive tape, overcoating the tape and wrap with electrical ignition waterproofing varnish.

5.2.1.1.7.2 Interunit cable assembly. This assembly shall be packaged in accordance with method IA-14 of Specification MIL-P-116.

5.2.1.1.7.3 Wiring systems and solenoids. Openings, exposed and unpainted surfaces of plugs, sockets, terminals, regulators, solenoids, wiring and wiring harness shall be covered with either pressure-sensitive tape or grade C greaseproof barrier-material, conforming to Specification MIL-B-121 and pressure-sensitive tape, overcoating the tape with electrical ignition waterproofing varnish. Loose ends or loops of wiring shall be secured in a protected location on the unit by either taping or tying.

5.2.2 Level C. The upper and lower cupola assemblies, including component parts thereto shall be cleaned and dried as specified in 5.1 and preserved and packaged in accordance with the manufacturer's commercial practice for immediate use.

5.3 Packing. Levels shall be as specified (see 6.1).

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5.3.1 Level A. The cupola assembly and components thereto processed as specified in 5.1 and 5.2 shall be packed in a type II, class 1 or 2, style a wood crate conforming to Specification MIL-C-104. The cupola assembly and its components shall be securely blocked, braced, or anchored to the skid base of the crate in accordance with the appendix to the crate specification and the following:

(a) The mounting frame fabricated in accordance with figure 1 shall be positioned on the skid base of the crate in such manner so that the sides of the mounting frame base, which contains the holes for anchoring the mounting frame to the skid base, shall be parallel to the wood skids. The mounting frame shall then be bolted to the skid base using six 5/8 inch carriage bolts with lock washers inserting one bolt in each end and one bolt in either one of the two center holes provided in the frame base. The bolts shall be of sufficient length to pass through the mounting frame base, the skid base flooring and wood sleepers. Two longitudinal chamfered wood sleepers not less than 4 feet long fabricated from nominal sized 2 by 4 lumber shall be added underneath the floor of the skid base at the bolt areas.

(b) The cupola assembly shall be placed in the mounting frame and bolted to the mounting frame top, using the same twelve bolts that are required to install the cupola assembly in the personnel carrier. Prior to use, these bolts shall be coated with type P-10, grade 2 or 3 preservative oil. After the bolts have been inserted and tightened, the exposed bolt surfaces shall be coated with type P-1 preservative compound.

(c) The turret shall be turned until the flame gun assembly is directed toward one of the mounting frame corners. The turret shall be held in this position by running steel strapping through the lifting eyes and around the mounting brackets of the interrupter ring. Four pieces of strapping shall be used on the three lifting eyes, and the strapping shall be done in such manner so that a criss-cross pattern will be obtained. The straps shall not be mounted to interrupter ring bracket nearest to a lifting eye. Flat steel strapping conforming to type I, class B, grade 2 of Specification QQ-S-781, size 3/4 inch wide by 0.35 inch thick shall be used. The traverse hand crank shall not be locked.

(d) The rotary joint shall have blocking and bracing applied in such manner as to prevent movement, turning or damage to the rotary joint or the central tubular column during shipment.

(e) The packaged interunit cable assembly (see 5.2.1.1.7.2) shall be secured to the skid base of the crate.

5.3.2 Level C. The cupola assembly and components thereto, processed as specified in 5.1 and 5.2 shall be packed in such manner as to insure carrier acceptance and safe delivery at first domestic destination for immediate use. Containers shall be in accordance with the Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

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5.4 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.

6.2 Caution. Adequate protection should be provided for all personnel when testing in accordance with this specification.

6.3 Test apparatus failure. If the test pressure cannot be maintained for a sufficient period of time in which to determine permanent volumetric expansion due to failure of the test apparatus, the test may be repeated at a pressure increase of 20 psig (when testing the gasoline tank).

6.4 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts. The repair parts packaging data sheets covering the applicable components of the cupola assembly should be used. (Copies of repair parts packaging data sheets may be obtained from the Director, Engineering and Industrial Services, ATTN: Chief, Design Branch, U. S. Army Edgewood Arsenal, Maryland, 21010.)

6.5 Supersession data. This specification supersedes Chemical Corps Purchase Description 196-131-756, dated 22 March 1962.

Custodian:

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Preparing activity:

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